Personalized Left Ventricle Reconstruction and Computational Blood Flow Dynamics in Hypertrophic Cardiomyopathy Patients

Owen Baenen[1]; Angie Carreño[2]; Sandra Rugonyi[1]; Theodore Abraham[2]
[1]Oregon Health and Science University, Portland, OR
[2] University of California San Francisco, Department of Medicine, Division of Cardiology, San Francisco, California

Introduction

Hypertrophic Cardiomyopathy **(HCM)** is a prevalent genetic heart condition characterized by abnormal ventricular septal hypertrophy and occasional left ventricular **(LV)** outflow obstruction. The hemodynamics of the LV plays a pivotal role in the functioning of the heart and evaluation of its performance; structural abnormality from HCM greatly influence the hearts contraction and relaxation patterns, invoke aberrant flow, and guide ventricular wall remodeling trajectories

Objective

We utilized Computational Fluid Dynamics **(CFD)** simulation to assess fluid/energy characteristics in patients with obstructive (**oHCM**), nonobstructive (**nHCM**) and healthy human hearts.

We also hoped to generate a pipeline for patient personalized image-based LV CFD simulation from a collection of single plane cardiac magnetic resonance images (cMRI).

Methods

HCM analysis with CFD was approached through two iterations.

1. The first pass had truncated models spanning the cardiac apex to the LV body before the valve partition. It included a model at end diastole and at end systole.
2. The second pass included models spanning the cardiac apex to both perivalvular regions with 20 timesteps of the whole cardiac cycle.

1. Custom semi-automated Python and MATLAB scripts reconstructed the 3D LV lumen geometry (Figure 1) from **cMRI** images of patients with (**oHCM**), (**nHCM**), and a healthy individual (n=1 each).
   - Mesh cross sections were generated through connection of 4 axial plane segmented rings and an apex point
   - Intermediate timesteps were linearly interpolated

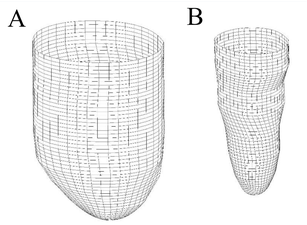


Figure 1. Example of corresponding meshes used to impose wall motion in LV models. **(A)** End diastole (ED), **(B)** End systole

2. A collection of single-plane cMRI series from the axial, sagittal, and coronal plane were segmented semi-automatically using SLICER 3D. Overlapping surface manifolds were processed and reconstructed with python PYVISTA library. Meshes were individually repaired and smoothed with the python MeshLab API pymeshlab.
   - Surface reconstruction of 20 timesteps was implemented
   - Corresponding time stepping meshes were generated through iterative vertex normal projection and Laplacian smoothing of a source mesh (time 1) to target manifolds.
   - Valve regions were clipped with suitable cutting planes

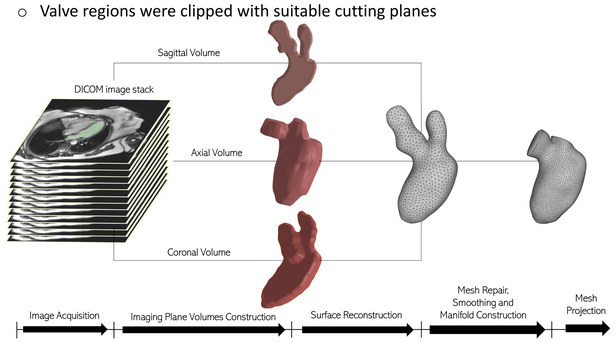


Figure 2. Example of the work flow for obtaining corresponding meshes from patient DICOM data

Results

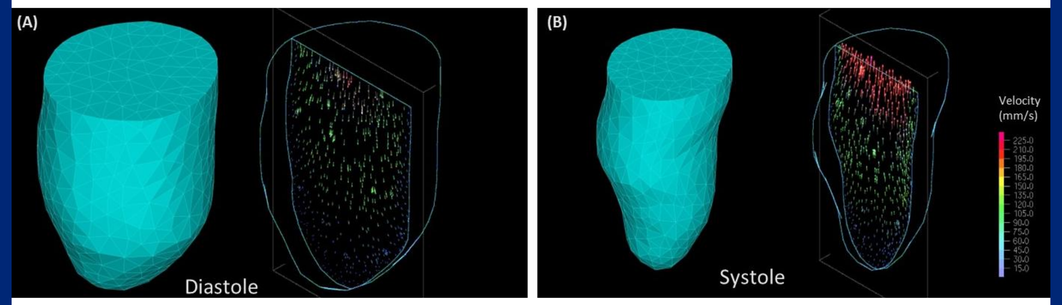
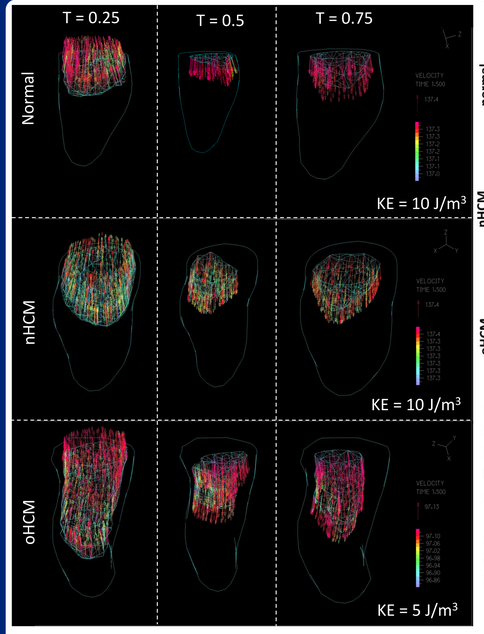


Figure 3: Patient-specific CFD modeling of HCM. 3D LV geometry and velocity vectors at a cut plane during diastole **(A)** and systole **(B)** for an nHCM patient.



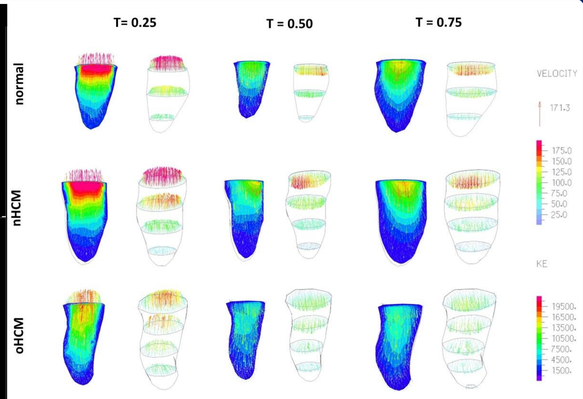
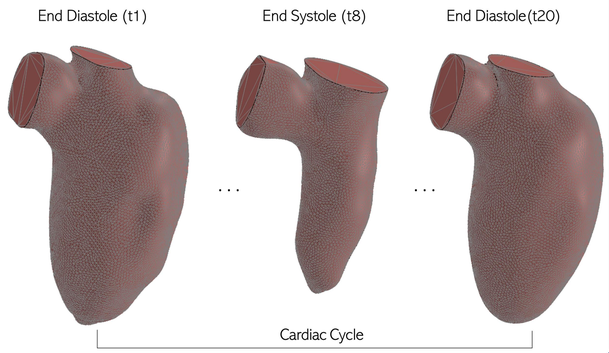


Figure 4 (above). Comparison of simulation results from CMR-based CFD modeling of the LV.
Patient specific CFD models of patient with oHCM, nHCM and healthy hearts. T represents 3 distinct phases during a cardiac cycle period. T = 0.25 corresponds to systole, while T=0.5 and T=0.75 correspond to diastolic filling. Blood velocities (mm/s) with colored arrows by magnitude are displayed in the short axis wireframe models. The flow kinetic energy (KE, in mJ/m³) profiles are shown as longitudinal cut planes for each model instance.

Figure 5 (above). Simulation of KE density and flow fields.
Simulation results are ordered identically to (Figure 4). Velocity vectors within the LV inside the volume of a KE isosurface (such that KE > isosurface value of 10 J/m³ or 5 J/m³). Colors represent magnitude and the outer layer is the whole LV.

Figure 6 (right). Simulation of KE density and flow fields.
Depiction of non-truncated LV manifold meshes. Models of 20 timesteps were generated. With meshes containing identical node count, uniform node correspondence, and identical face connectivity, displacements can be applied strategically to simulate the motion of the LV.



Results & Discussion

3D Simulation results (Figure 3, 4) confirmed **similar stroke volume** for the normal and nHCM patients (45 ml and 59 ml, respectively), and **reduced stroke volume** (34 ml) for the oHCM patient

Energy flow rate from the LV in oHCM is **significantly reduced**; the maximum exiting flow energy rate during systole is about 3 mW for both the normal and nHCM patients, and about 0.5 mW for the oHCM patient.

KE and blood Flow velocities are reduced in oHCM (Figure 4). There is noticeable stroke volume differences between oHCM and the healthy heart and nHCM patient. The reduced stroke volume and asymmetric fluid flow in/out of the LV during diastole/systole (isosurface analysis in Figure 5) implies **inefficient LV energy transfer** in the case of oHCM and partially in nHCM. Reduced KE can be attributed to loss through inefficient LV contraction/relaxation from the structural abnormality of nHCM and oHCM

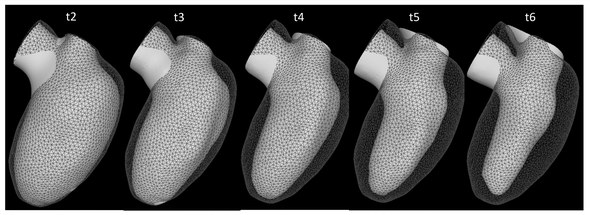


Figure 7: Non-truncated LV Model Example (nHCM). Example sequence of consecutive 3D mesh manifolds (white) compared to t1 (wireframe) during systole (20 timesteps in full cardiac cycle). Mesh correspondence was generated through projection of source mesh (t1) iteratively onto subsequent meshes.

Our first pass CFD modelling analysis of HCM in human LV contained only a small subset of cardiac cycle phases. It was also truncated below the valves and intermediate phases were linearly interpolated yet they **revealed significant trends** in flow patterns between nHCM, oHCM, and normal heart.

CFD modelling of HCM with increased surface definition, peri-valvular morphology, and time steps will offer **a more complete image of HCM** vorticity, KE, and flow patterns. 3D LV modelling is still limited by non-automatic segmentation and reconstruction; more study might better characterize how asymmetric fluid flow and morphology in HCM influences cardiac performance.

Conclusion

Patient-specific, image-based CFD models capture disruption of LV blood flow dynamics and energy profiles in HCM sub-types.
CFD analysis of LV time stepping manifolds can characterize the interplay between structural abnormality, aberrant fluid flow profiles, to cardiac performance.
CFD analysis in patient LV can provide insights into cardiac wall remodeling trajectories in HCM and noninvasive evaluation of cardiac performance.

Acknowledgements

This work was supported by NIH grants and the OHSU ORION program through the American Heart Association. Imaging data was made available thanks to Angie Carreño and Theodore Abraham from University of California San Francisco, Department of Medicine Division of Cardiology, San Francisco, California. A special thank you to my mentor Sandra Rugonyi, and others for their guidance.